CLEARPATH

REGULATORY READINESS REPORT · TIER 1

ASSESSMENT 0BE5A3DB

REGULATORY SCORECARD

Bioresorbable Cardiac Stent

Class D · novel · MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

LIKELY CLASSIFICATION

Class D · novel

High complexity

CONFIDENCE

High confidence

LIKELY PATHWAY

MD-7 (Central) · likely MD-22 / test-licence (MD-13) path

Clinical investigation likely required.

TIMELINE + COST BAND

36–60 months

Likely compliance spend: ₹17–41 L

READINESS

1 / 10

Band: Red (significant work ahead)

RISK LEVEL

High

Risk = potential clinical/regulatory exposure today.

What we assumed about your device

CORRECT IN EDITOR

We inferred the fields below from your wizard answers and uploaded materials. Correct anything that's wrong — the Submission Pack regenerates from your corrected values.

ESTIMATED

Information role — Diagnoses/treats

Inferred from device description — a stent physically treats coronary artery disease. Hardware founders don't answer this directly.

↳ CORRECT AT EDITOR OR BY CHANGING YOUR ONE-LINER

ASSUMED

Year-1 users — Under 10,000

Hardware launches typically stay below 10 lakh users in Year 1. Correct in your editor if you expect scale that triggers DPDP SDF designation.

↳ CORRECT AT EDITOR

ESTIMATED

Sterile device — Yes (assumed)

Inferred from patient-contact tier — implant placed via catheter into coronary arteries must be supplied sterile.

↳ CORRECT AT EDITOR §14

ASSUMED

Drug content — No (assumed)

One-liner describes a bioresorbable stent without a drug-elution claim. Correct in your editor if your stent is drug-eluting — that would shift documentation and may invoke combination-product review.

↳ CORRECT AT EDITOR

ASSUMED

Ionising radiation — No (assumed)

Stents do not emit ionising radiation. Correct if your device incorporates radioactive markers.

↳ CORRECT AT EDITOR

ASSUMED

Intended population — Humans only

Assumed humans-only based on coronary artery disease indication.

↳ CORRECT AT EDITOR

ASSUMED

Manufacturing location — India (assumed)

No facility details provided; defaulted to India. Correct if you plan to manufacture abroad and import (MD-14 path).

↳ CORRECT AT EDITOR

ESTIMATED

Measuring function — No

A stent provides structural support and does not measure physiological parameters.

↳ CORRECT AT EDITOR

ESTIMATED

CDSCO class — D

Class D — derived from implant_gt_30d patient contact (long-term implant). Long-term implants automatically classify as Class D under MDR 2017 Fourth Schedule.

↳ CORRECT AT EDITOR §4

EXTRACTED

Implantable — Yes

Directly from Q9 = implant_gt_30d and the one-liner ('implanted via catheter, dissolves over 24 months').

↳ CORRECT AT EDITOR

ESTIMATED

Software in device — No

No mention of firmware, algorithm, or software in the device description. Bioresorbable stent is a passive mechanical implant.

↳ CORRECT AT EDITOR

WHAT TRIGGERED THIS CLASSIFICATION

- Influences clinical management
- Novel indication (no Indian predicate)

TOP GAPS TO CLOSE

- ISO 10993 biocompatibility evidence not documented
- No predicate + no clinical data — MD-26/MD-27 pre-permission and pivotal investigation needed
- ISO 13485 QMS not detected

RECOMMENDED NEXT ACTION

Scope an ISO 10993 testing plan for a long-term blood-contacting implant — minimally -4 (blood interactions), -6 (implantation), -10 (sensitisation), -11 (systemic toxicity), plus degradation/leachables for the bioresorbable polymer. Engage a NABL-accredited testing lab.

SECTION 2

Your Likely Regulatory Pathway

Why this likely class applies to your product — and the form sequence reviewers typically expect.

A bioresorbable cardiac stent is a long-term implant (greater than 30 days) placed in direct, sustained contact with the central circulatory system, which is among the most sensitive clinical settings recognised under Indian medical device rules. The device actively supports vessel patency and resorbs over time, so any failure mode may directly affect cardiac function rather than simply informing a clinician's decision. Based on published CDSCO guidance, this combination of life-sustaining function, critical anatomical site, and novel resorbable behaviour typically places the product in the highest risk tier. With no established Indian predicate, the pathway likely runs as a clinical investigation followed by MD-7 manufacturing licensure through the Central Licensing Authority.

AUTHORITY **CDSCO Central Licensing Authority** FORMS **MD-26** **MD-27** **MD-12** **MD-13** **MD-22** **MD-7** **MD-9**

LIKELY STEP SEQUENCE

01 MD-26 / MD-27 pre-permission

File the pre-permission application before MD-7 because no Indian predicate is claimed. Adds a separate review cycle.

2–4 MONTHS

02 Test licence (MD-12 / MD-13)

Apply via MD-12 for a test licence (granted as MD-13) so a small number of devices can be made for the clinical investigation.

2–3 MONTHS

> **TEST LICENCE NOTE**
>
> A test licence (granted as MD-13, applied via MD-12) typically precedes the clinical investigation. The investigation itself runs under MD-22 / MD-23 permissions before the device can be cleared for commercial manufacture.

03 MD-22 clinical-investigation permission + EC approval

Design the multi-centre study, get EC clearance, register on CTRI, enrol participants.

9–14 MONTHS

04 MD-7 manufacturing licence (CLA → MD-9)

File MD-7 with the Central Licensing Authority alongside DMF, PMF, Fifth-Schedule QMS, clinical evidence, biocompatibility (and sterilization validation where applicable). CDSCO MD Officer team conducts a pre-grant site audit within 60 days of application.

6–9 MONTHS

SECTION 3

Readiness Gap Analysis

The gaps a reviewer is likely to flag, ordered by priority. Effort and cost bands are typical Indian medtech ranges — your numbers may be lower with in-house resources or higher with a Big-Four-style consultant.

PRI	GAP	WHY IT MATTERS	SUGGESTED NEXT STEP	EFFORT · COST
P1	ISO 10993 biocompatibility evidence not documented	A bioresorbable cardiac stent sits in long-term blood and tissue contact, so reviewers will likely expect a full ISO 10993 evidence package covering ISO 10993-4 (blood contact), -6 (implantation), -11 (systemic toxicity), and degradation-product testing under -13/-17 given the resorbable polymer chemistry. For Class D implants this is one of the first items reviewers open in DMF Section 8.11, and without test reports from a recognised lab the file typically cannot move past initial screening.	Scope an ISO 10993 testing plan for a long-term blood-contacting implant — minimally -4 (blood interactions), -6 (implantation), -10 (sensitisation), -11 (systemic toxicity), plus degradation/leachables for the bioresorbable polymer. Engage a NABL-accredited testing lab.	3–6 months · ₹2–8 L
P1	No predicate + no clinical data — MD-26/MD-27 pre-permission and pivotal investigation needed	Because this is a novel Class D implant with no clear predicate, reviewers will likely require a CDSCO pre-permission via MD-26 followed by a pivotal MD-27 clinical investigation in Indian patients, with Ethics Committee approvals and CTRI registration in place before first subject enrolment. Based on published guidance this is typically the longest item on the critical path — often 18-30 months — so any delay in protocol design, site selection, or DCGI interaction tends to set the overall launch date.	Plan a CDSCO pre-submission meeting to align on pre-permission requirements (MD-26/MD-27) and pivotal clinical investigation design (MD-22). Budget for first-in-human and pivotal trials.	9–14 months · ₹8–18 L
P1	ISO 13485 QMS not detected	For a Class D cardiovascular implant, reviewers will likely expect a mature ISO 13485 quality system already operating — design controls, CAPA, supplier qualification, internal audits, and traceable batch records — not a system being built in parallel with the submission. Without a certificate (or strong evidence of a certification audit in progress), the manufacturing-licence application is typically held at intake.	Engage an ISO 13485 consultant for a gap assessment and begin building a design-control and risk-management (ISO 14971) framework appropriate for a Class D implantable.	6–9 months · ₹3–5 L

P2	Sterilization validation method not documented	Reviewers will likely expect a validated terminal sterilization process matched to the resorbable polymer's tolerance (radiation per ISO 11137 or EO per ISO 11135 are typical for stents), with bioburden data, SAL 10^-6 demonstration, dose-setting or half-cycle studies, and packaging-integrity and shelf-life validation referenced in DMF Section 8.14. Missing or partial validation is one of the more common reasons Class C/D files come back with queries.	Pick the method (EtO / gamma / steam / aseptic), engage a validated sterilization partner, and assemble the validation file per the relevant standard (ISO 11135 / 11137 / 17665).	2–4 months · ₹1.5–3 L
P3	Device Master File (Appendix II) not assembled	The Device Master File under Appendix II of the Fourth Schedule is the document reviewers read end-to-end, and every other claim in the application is cross-checked against it — design inputs, risk file, V&V, biocompatibility, sterilization, stability, clinical evidence, PMS plan, and batch CoA. For a novel Class D device, an incomplete or loosely assembled DMF typically triggers multiple rounds of queries and can stall the file for months even when the underlying testing exists.	Assemble the DMF per Appendix II of the Fourth Schedule — descriptive info, classification, product specification, predicate comparison, labelling, design & mfg, essential principles, risk file, V&V, biocomp, sterilization, stability, clinical evidence, PMS, batch CoA.	2–4 months · ₹1–3 L

SECTION 4

Timeline + Cost Estimator

Phased roadmap with Indian-context cost bands. Total range and anchor reflect the readiness card's headline estimate.

TOTAL **36–60 months** Class D novel implantable: MD-26/MD-27 pre-permission, pivotal clinical investigation (MD-22), manufacturing li and post-market surveillance — typically a 3–5 year arc from pre-MVP.

Phase 1 — Foundational compliance

3–6 MONTHS **₹3–6 L**

Engage an ISO 13485 consultant for a gap assessment, document the Intended Use Statement, and lock the classification rationale.

Phase 2 — Technical documentation

4–7 MONTHS **₹3–9 L**

Build the Device Master File (Appendix II) — descriptive info, predicate comparison, design & manufacturing, essential principles, ISO 14971 risk file; ISO 10993 biocompatibility evidence at the tier matching patient contact; Sterilization validation per the chosen method (ISO 11135 EO / ISO 11137 radiation / ISO 17665 steam).

Phase 3 — Clinical validation

9–14 MONTHS **₹8–18 L**

Design and run a multi-centre clinical investigation with EC approval and CTRI registration. Typically the longest item on the critical path.

Phase 4 — Submission preparation

2–3 MONTHS **₹2–5 L**

Assemble the Plant Master File (Appendix I), finalise the Device Master File, labelling and IFU, predicate or substantial-equivalence narrative, and pre-submission review.

Phase 5 — CDSCO review cycle

4–8 MONTHS **₹1–3 L**

Submission filing (MD-3 to SLA or MD-7 to CLA depending on class), queries-and-responses cycle, pre-grant site audit (60 days for Class C/D, 90 days for Class B), licence grant.

LIKELY BOTTLENECKS

- Clinical investigation timeline (EC + CTRI + enrolment + follow-up) dominates the schedule.
- Top high-severity gap: ISO 10993 biocompatibility evidence not documented.
- Novel device path adds MD-26 / MD-27 pre-permission ahead of MD-7.

SECTION 5

Reviewer Insights

What CDSCO reviewers will likely look for in a submission of this shape. Generic regulatory checklists rarely capture reviewer psychology — this is yours.

Strength of clinical validation evidence

For a novel Class D bioresorbable cardiac stent, reviewers will likely expect a prospective Indian clinical investigation with EC approvals, CTRI registration, and a pre-specified SAP covering primary endpoints like target lesion failure, late lumen loss, and scaffold thrombosis. Resorption kinetics, imaging follow-up (OCT/IVUS) timelines, MACE adjudication, and sample-size justification typically face line-by-line scrutiny, alongside long-term follow-up commitments extending past full bioresorption.

Intended Use consistency across documents

Reviewers typically cross-check the Intended Use Statement against IFU, labelling, IB, and marketing material for the stent. Drift around lesion type (de novo vs restenotic), vessel diameter range, patient population (e.g., diabetic subsets), or claims about bioresorption benefits over DES may trigger early queries. Implant indications, contraindications, and operator-training requirements should read identically across the dossier.

Biocompatibility tier matching patient contact

As a long-term blood-contacting implant, reviewers will likely expect a full ISO 10993-1 risk assessment covering 10993-4 (haemocompatibility, thrombogenicity), 10993-6 (sub-chronic and chronic implantation), 10993-10 (sensitisation, irritation), 10993-11 (systemic and sub-chronic toxicity), and 10993-17/-18 (leachables/extractables across the resorption profile). Degradation by-product characterisation and chemistry across the resorption timeline typically receives particular attention.

Indian-population validity

Given novel status, reviewers may probe how scaffold performance translates to Indian CAD patients — smaller mean reference vessel diameters, higher diabetic and young-MI prevalence, and differing lesion morphology. Expect questions on subgroup analyses by vessel size, diabetes status, and lesion complexity, plus how cath-lab imaging protocols (OCT/IVUS availability) and post-procedural DAPT adherence patterns were accounted for in the Indian arm.

Cybersecurity & data handling

Cybersecurity is typically lower-weight for a passive implant, but if the deployment system, procedural planning software, or post-market registry collects PHI or imaging, reviewers may ask for an IEC 81001-5-1 threat model, encryption at rest and in transit, role-based authentication, breach notification SOP, and CERT-In safe-to-host evidence for any cloud-hosted patient registry or follow-up imaging repository.

SECTION 6

Smart Examples

Annotated good-vs-bad wording. These are snippets, not full forms — copy them as models for your own Intended Use, claim, and risk-justification language.

INTENDED USE Concrete user + setting + function

GOOD

Intended for implantation by qualified interventional cardiologists in cath-lab settings for the treatment of de novo native coronary artery lesions in adult patients eligible for percutaneous coronary intervention. The scaffold provides temporary vessel support and drug elution, then resorbs over the documented timeframe; clinical decision-making, lesion selection, and post-procedure management remain with the treating cardiologist.

BAD

A next-generation bioresorbable stent that helps patients recover faster and restores natural artery function without leaving anything behind.

WHY THIS IS SAFER

The first wording names the user, setting, patient group, lesion type, and the scaffold's bounded mechanical role — all verifiable against the IFU. The second is marketing copy that overstates outcomes and invites reviewer questions on evidence for 'natural function' and 'faster recovery'.

CLAIM WORDING Indication scope vs. label creep

GOOD

Indicated for adult patients (18+) with de novo native coronary artery lesions of reference vessel diameter 2.5-3.5 mm and lesion length up to 24 mm, treated by elective PCI. Not validated for use in left main disease, bifurcation lesions requiring two-stent strategy, in-stent restenosis, bypass grafts, STEMI within 24 hours, paediatric patients, or vessels with heavy calcification beyond what was represented in the pivotal cohort.

BAD

Suitable for treatment of coronary artery disease and restoration of vessel patency in adult patients.

WHY THIS IS SAFER

The first wording fixes the population, vessel dimensions, lesion type, and explicit exclusions a reviewer can map to the clinical investigation cohort. The second is open-ended and invites label-creep questions on left main, bifurcation, and acute MI use that the evidence likely does not cover.

RISK JUSTIFICATION

Residual risk acceptance

GOOD

After applying the documented controls (polymer characterisation, scaffold thrombosis mitigation via DAPT labelling, strut-fracture testing per ISO 25539-2, resorption-kinetics studies), the residual risk of late scaffold thrombosis is judged acceptable in light of the clinical benefit demonstrated in the pivotal investigation (TLF and device-oriented endpoints at 12 and 24 months). Residual risk is monitored under the PMCF plan with five-year imaging follow-up and quarterly review of MDR-reportable events.

BAD

Residual risks including thrombosis and resorption-related events have been reviewed and accepted as low.

WHY THIS IS SAFER

The first wording ties acceptance to named controls, specific endpoints, and a PMCF mechanism — what ISO 14971 Clause 8 and MEDDEV-style review expect for a Class D implant. The second is a bare assertion with no evidence anchor and no surveillance commitment for late events.